Ho Chi Minh City University of Technology



Faculty of Computer Science and Engineering
Course: Computer Architecture Lab (CO2008)

Lab 2

Branches and procedures

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1 Introduction

- This report presents the solutions for Lab 2 – *Branches and procedures*.
- All programs are written in MIPS assembly and tested on the MARS MIPS simulator.
- The programs use conditional branch and unconditional jump instructions, and call procedures. Exercises 2 and 4 are solved with recursive procedures.
- Each exercise is verified with three test cases. For every test case, the input, the expected output and a screenshot of the actual run are given.

2 Exercises

2.1 Exercise 1

Requirement

Write a MIPS program with the following requirements:

1. Declare an integer array of width 5 with data entered from the keyboard.
2. Check whether each element in the array is divisible by 4 or not.
3. If an element is not divisible by 4, change it to the number divisible by 4 that is closest to it (round down is prioritized). For example, if the number is 35, the result will be 36, if the number is 34 or 33, the result will be 32.
4. Print the result to the terminal.

Test cases

Test case 1

- **Input:** `35, 34, 33, 32, 31`
- **Expected output:** `Result: 36 32 32 32 32`
- **Output:**

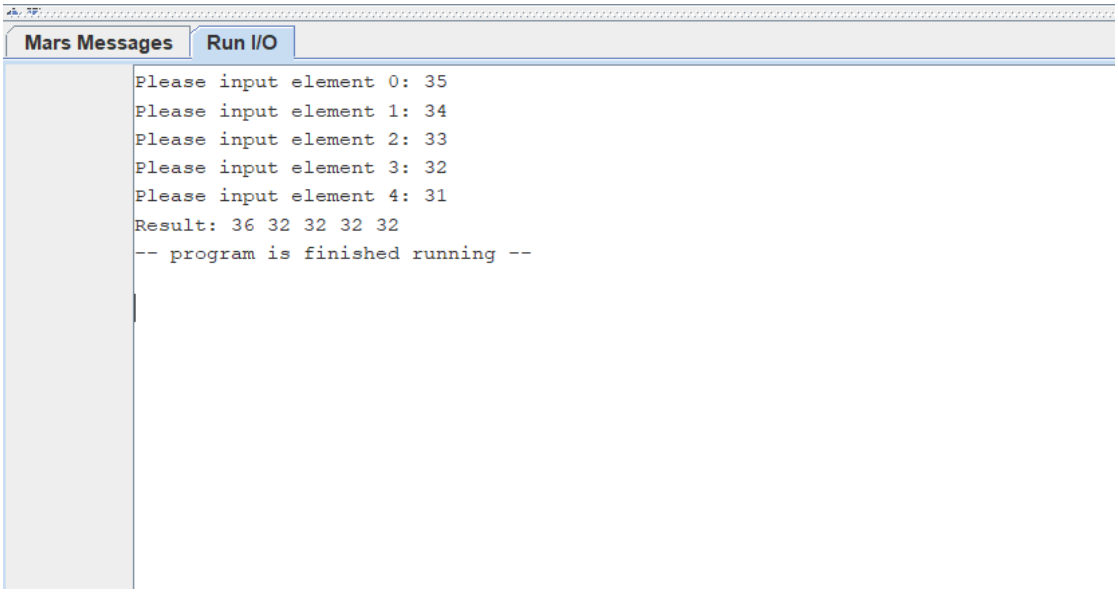


Figure 1: Exercise 1 – Test case 1

Test case 2

- **Input:** `0, 1, 2, 3, 4`
- **Expected output:** `Result: 0 0 0 4 4`
- **Output:**

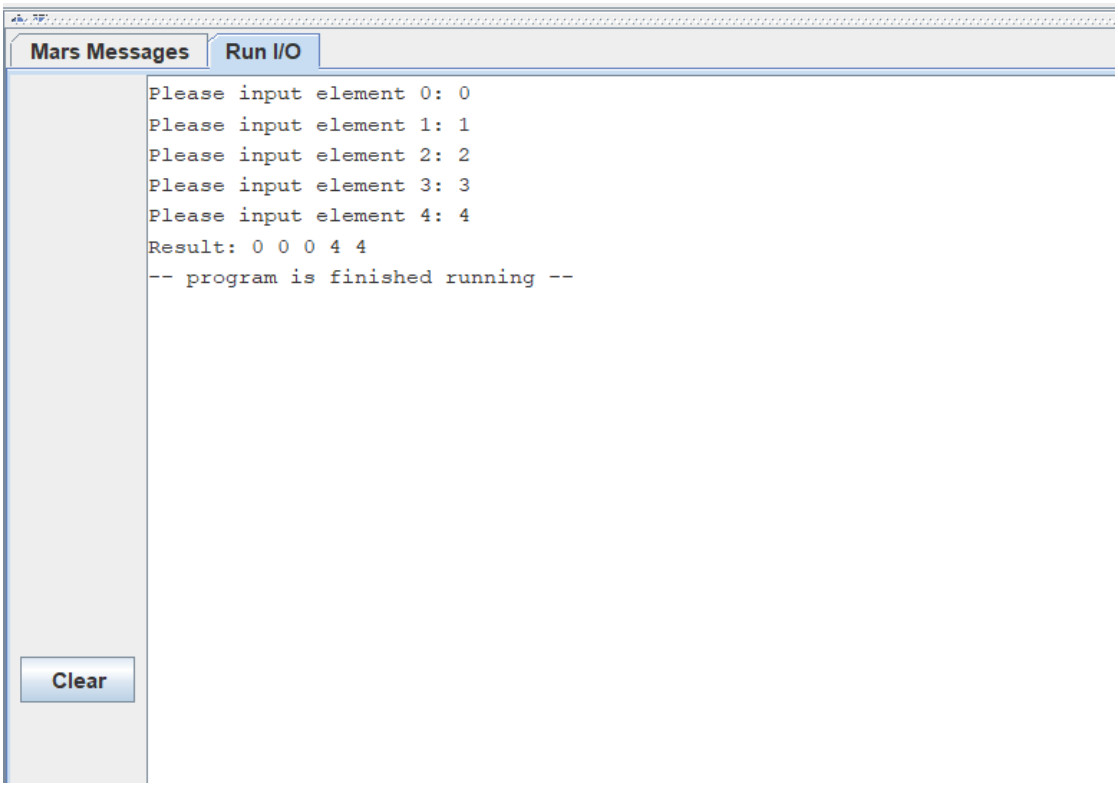


Figure 2: Exercise 1 – Test case 2

Test case 3

- **Input:** `-1, -2, -3, -5, 100`
- **Expected output:** `Result: 0 -4 -4 -4 100`
- **Output:**

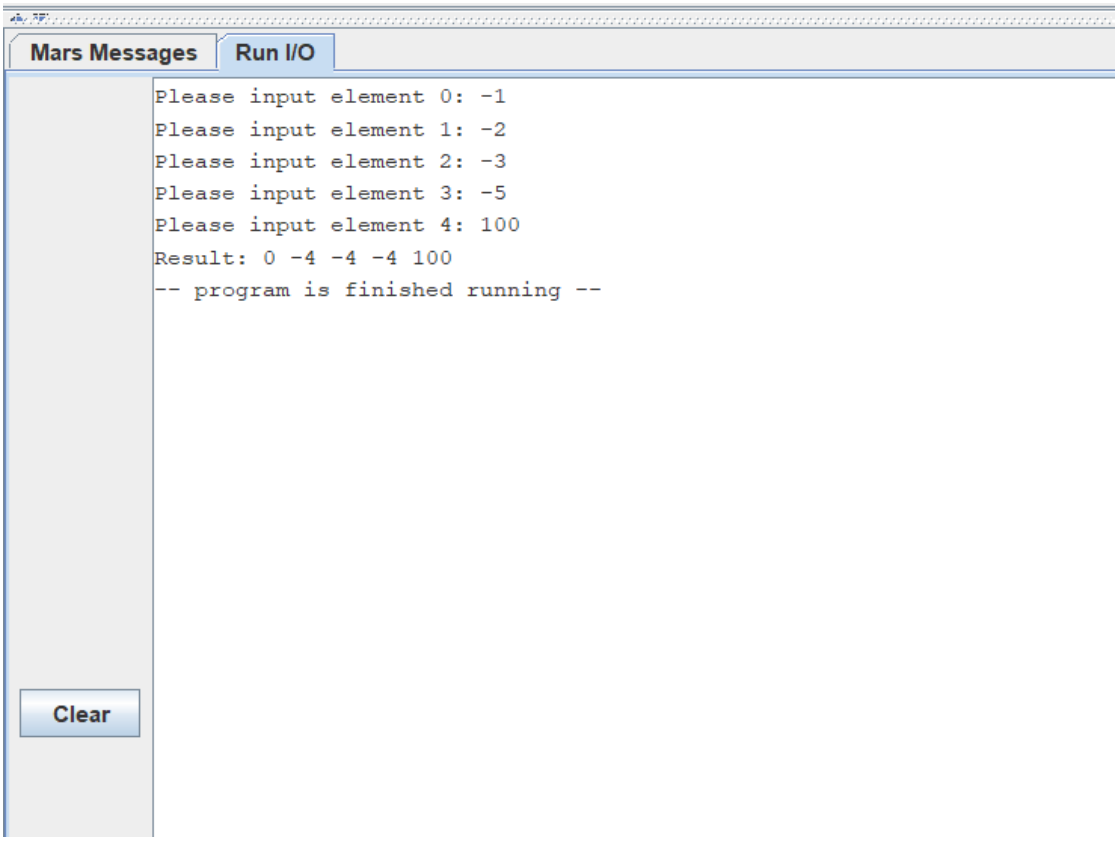


Figure 3: Exercise 1 – Test case 3

2.2 Exercise 2

Requirement

Write a MIPS program that requests two positive integers called a and b from the user (please check if they are positive or not). The program then prints their greatest common divisor (GCD) and lowest common multiple (LCM) utilizing recursive. The input numbers should be after the prompt: "a = " and "b = ". For example, if the user input **a=6** and **b=45**, the result should be **GCD=3, LCM=90**.

Test cases

Test case 1

- **Input:** $a = 6, b = 45$
- **Expected output:** `GCD=3, LCM=90`
- **Output:**

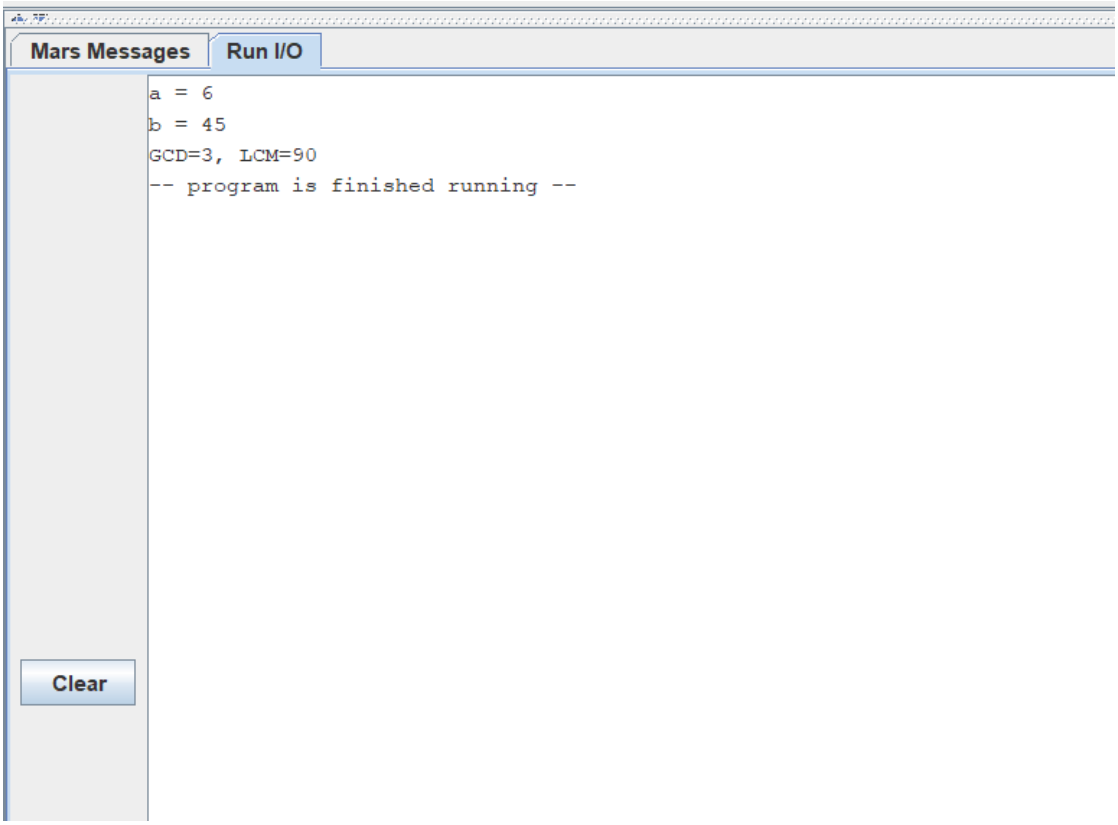


Figure 4: Exercise 2 – Test case 1

Test case 2

- **Input:** $a = 12$, $b = 18$
- **Expected output:** `GCD=6, LCM=36`
- **Output:**

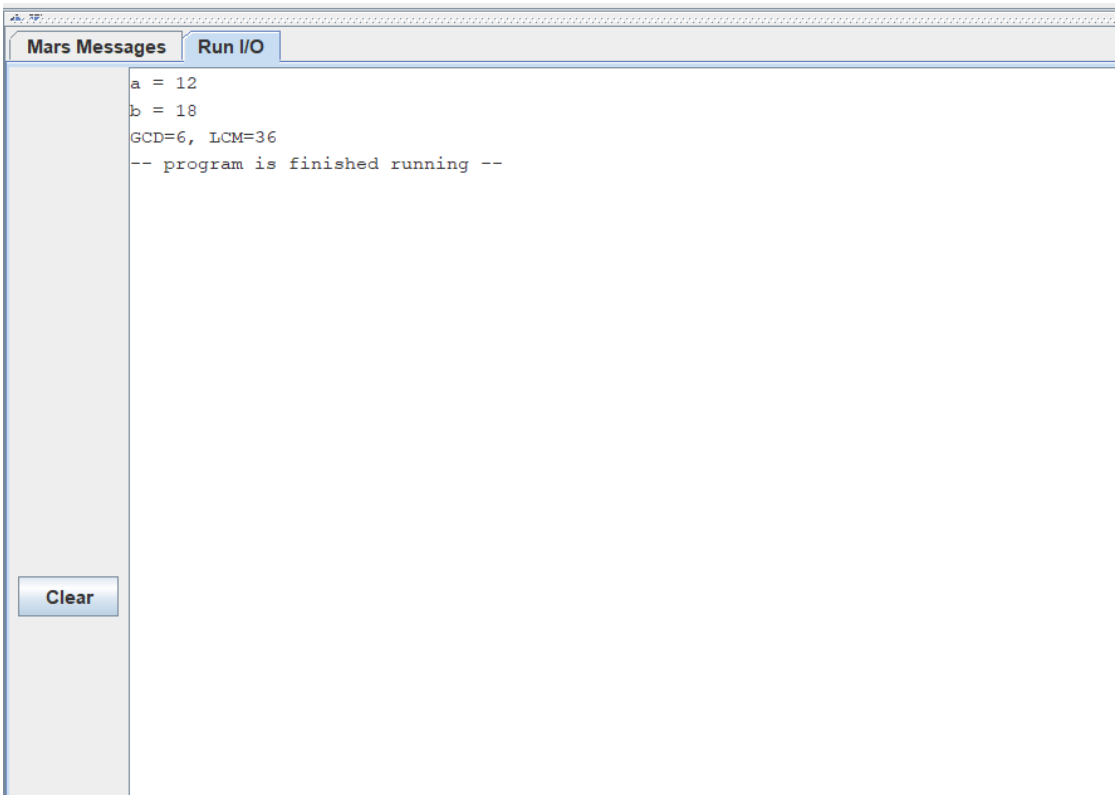


Figure 5: Exercise 2 – Test case 2

Test case 3

- **Input:** $a = -4$ (rejected), $a = 8$, $b = 0$ (rejected), $b = 12$
- **Expected output:** two messages `Invalid input. Please enter a positive integer.`, then `GCD=4, LCM=24`
- **Output:**

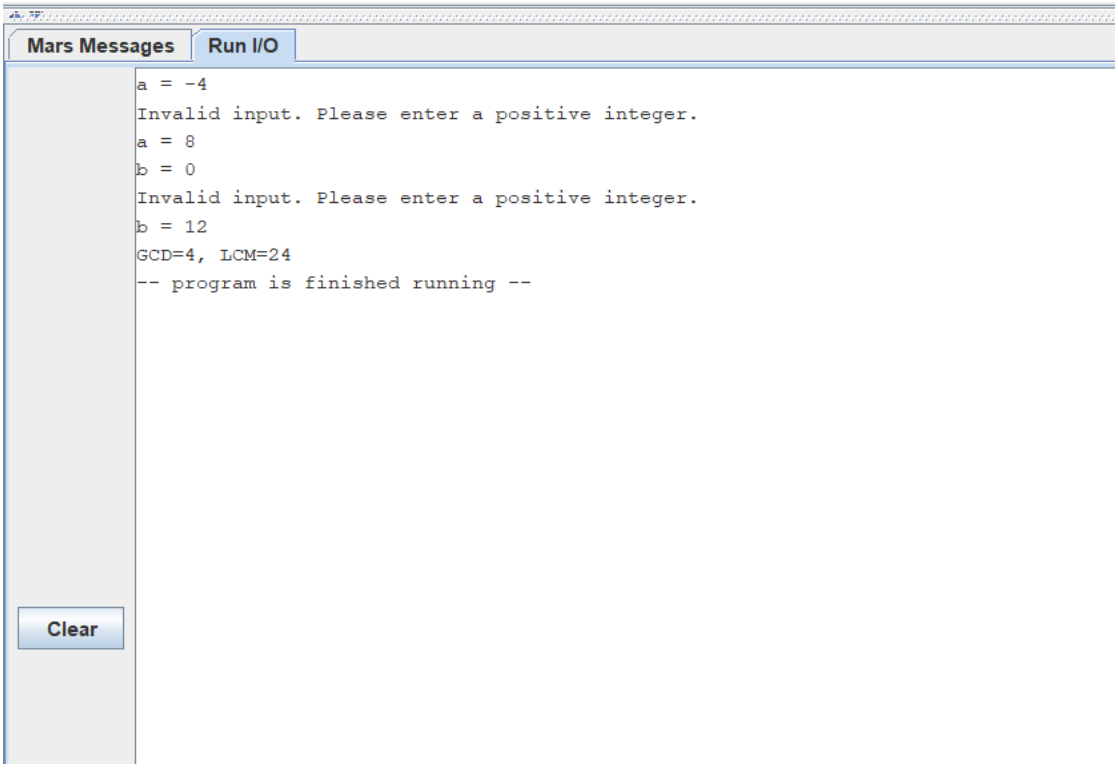


Figure 6: Exercise 2 – Test case 3

2.3 Exercise 3

Requirement

Write a MIPS program to find the second largest element in a 10-elements integer array. The program is also required to find all their indexes. Print the value and all of its indexes. The elements of the array must be inserted by the user after the prompt: "Please insert a element:"

For example, if the array is **1, 2, 7, -7, 3, 7, 4, 6, 8, 7** the output should be **Second largest value is 7, found in index 2, 5, 9**.

Test cases

Test case 1

- **Input:** `1, 2, 7, -7, 3, 7, 4, 6, 8, 7`
- **Expected output:** `Second largest value is 7, found in index 2, 5, 9`
- **Output:**

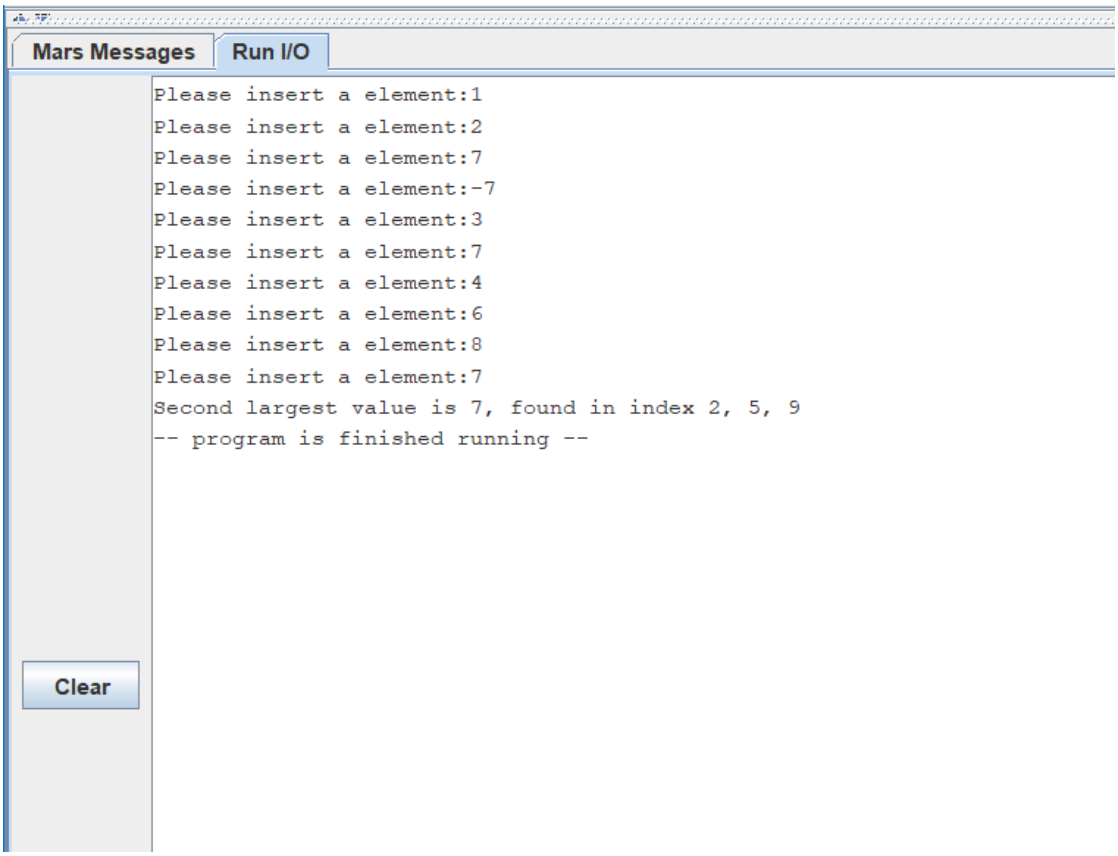


Figure 7: Exercise 3 – Test case 1

Test case 2

- **Input:** `10, 9, 8, 7, 6, 5, 4, 3, 2, 1`
- **Expected output:** `Second largest value is 9, found in index 1`
- **Output:**

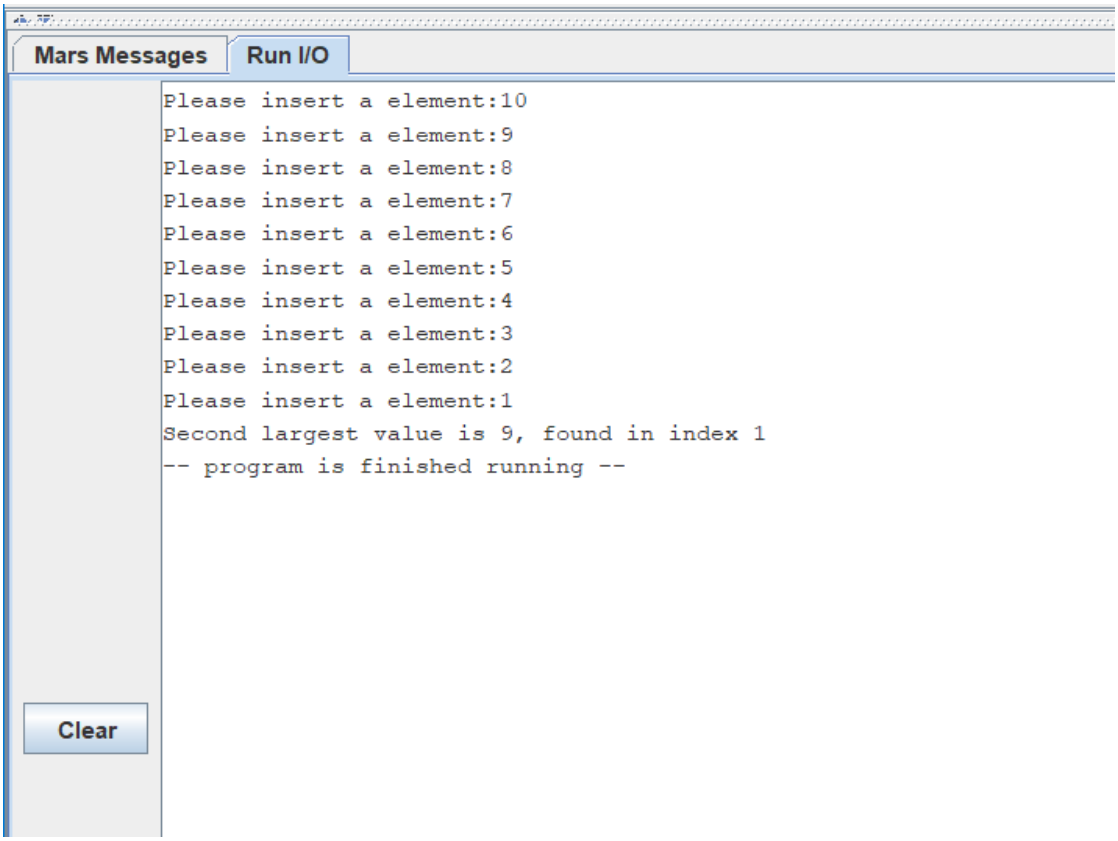


Figure 8: Exercise 3 – Test case 2

Test case 3

- **Input:** `4, 4, 4, 4, 4, 4, 4, 4, 4, 4`
- **Expected output:** `There is no second largest value.`
- **Output:**

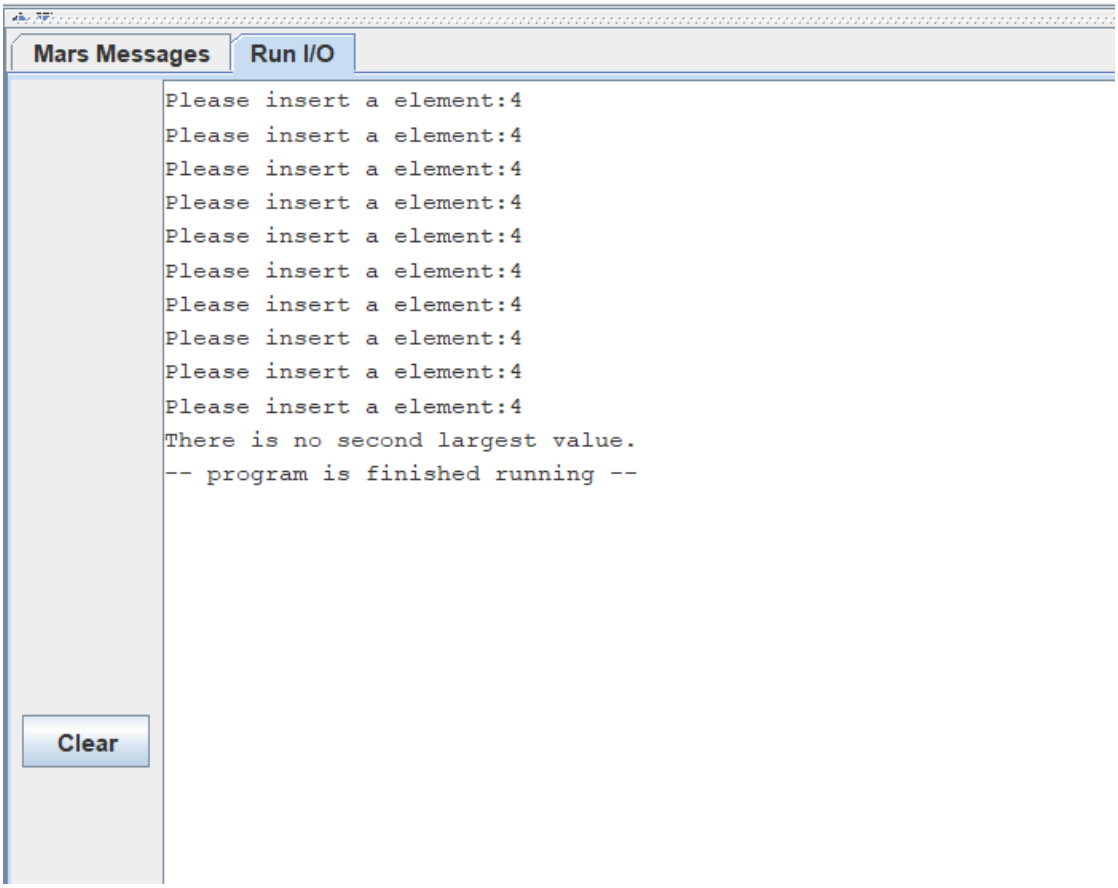


Figure 9: Exercise 3 – Test case 3

2.4 Exercise 4

Requirement

Write a MIPS program to take a non-negative integer from the user (you need to check if the user input the correct number). Print the factorial number of that integer into the terminal.
For example, if the user input **5**, the output should be **120**. If the user input **-2**, the output should be **A factorial for your number can not be found.**

Test cases

Test case 1

- **Input:** `5`
- **Expected output:** `120`
- **Output:**

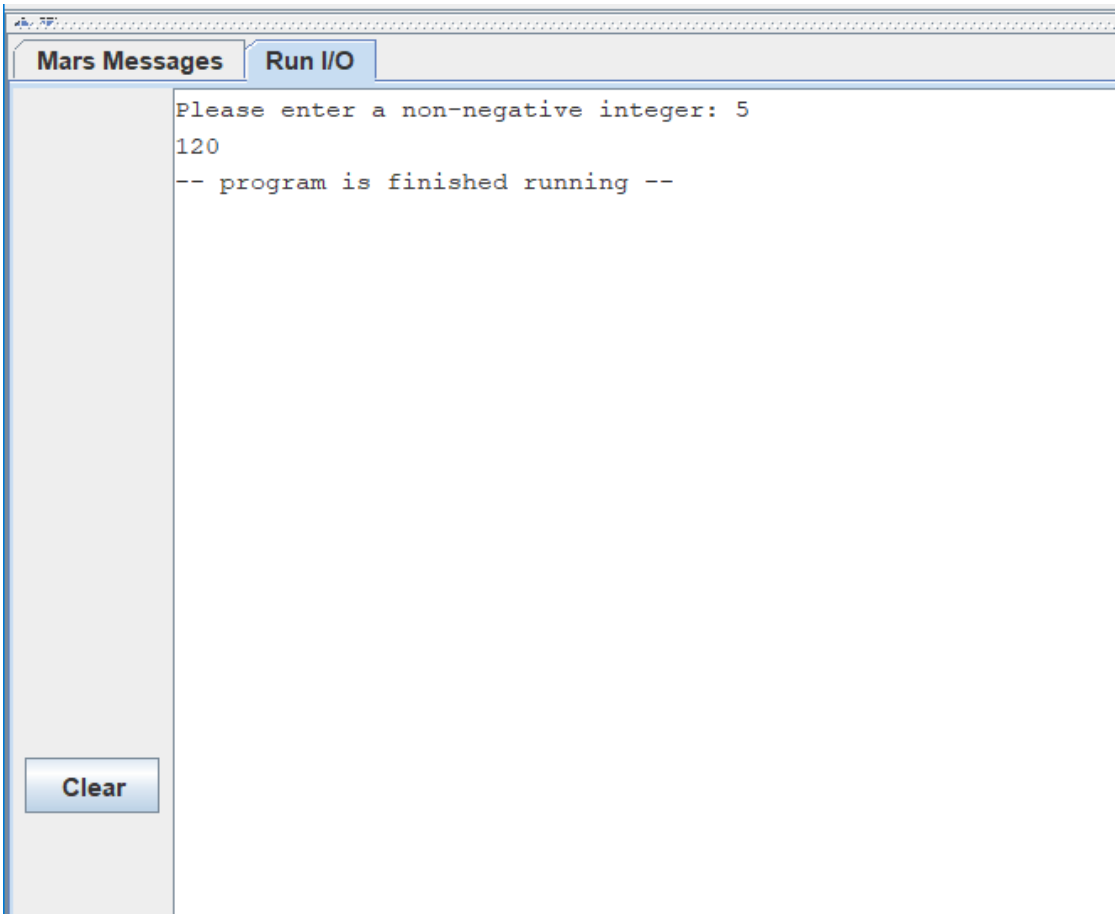


Figure 10: Exercise 4 – Test case 1

Test case 2

- **Input:** `0`
- **Expected output:** `1`
- **Output:**

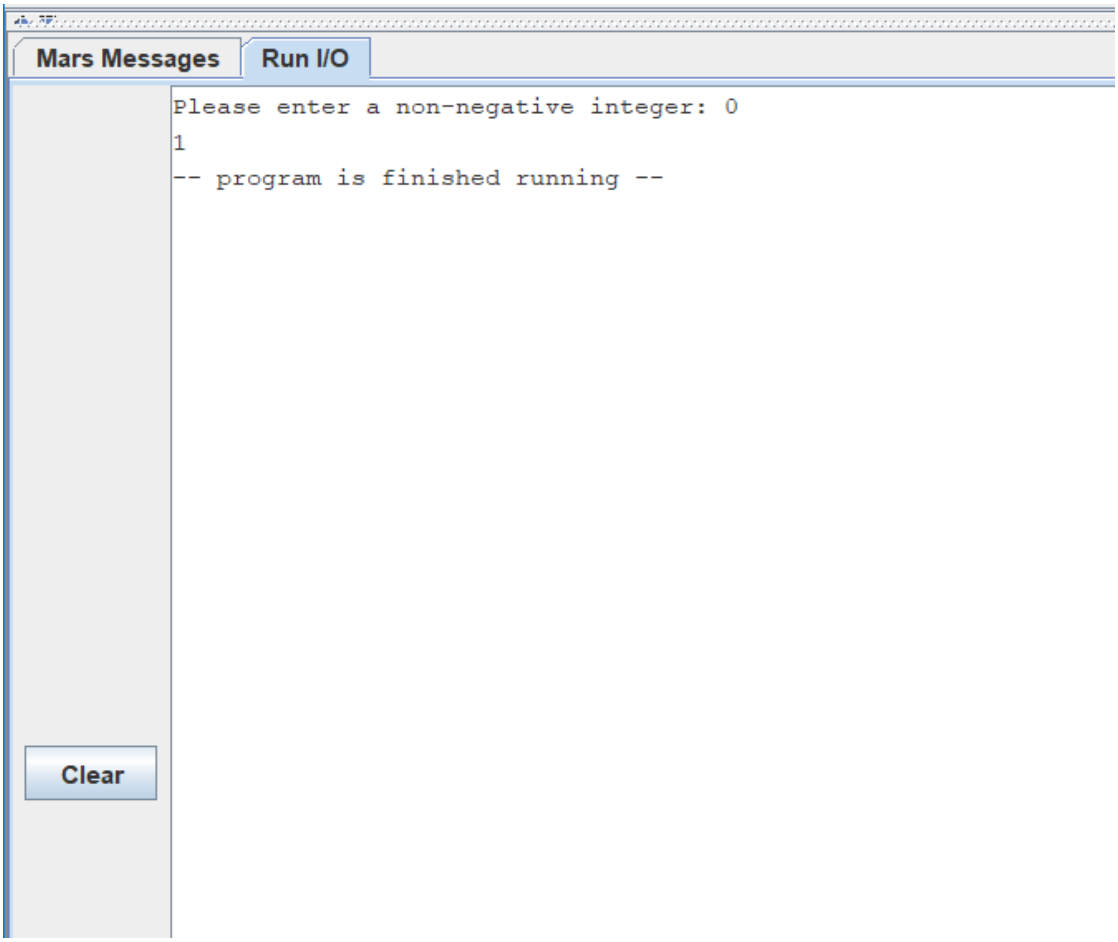


Figure 11: Exercise 4 – Test case 2

Test case 3

- **Input:** `-2`
- **Expected output:** `A factorial for your number can not be found.`
- **Output:**

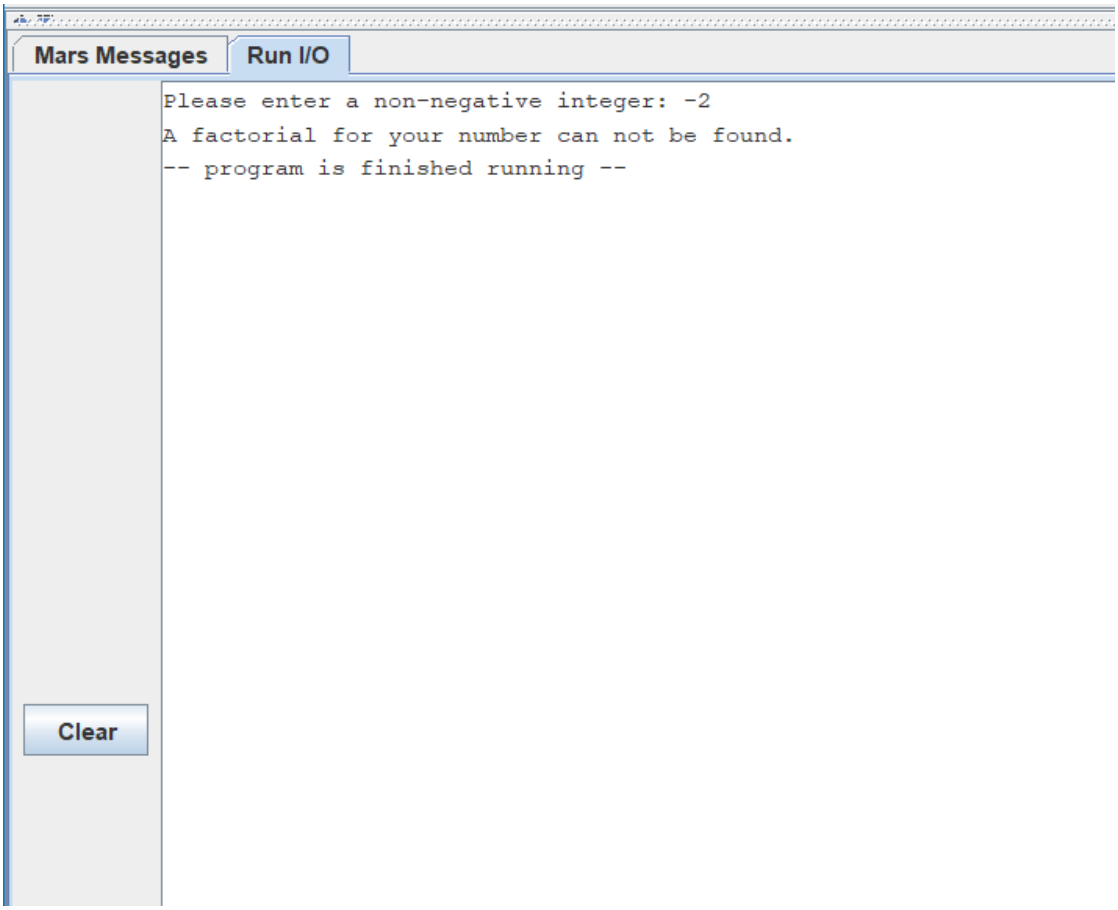


Figure 12: Exercise 4 – Test case 3